C. Dungeon

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, brute force, data structures, greedy, sortings |
https://codeforces.com/problemset/problem/2164/C

You are now in a dungeon with n swords, facing m monsters.

The damage of the i-th sword is a_i, and the life value of the i-th monster is b_i. A sword with damage x can kill a monster with life value y if and only if $x \geq y$.

After killing the i-th monster with a sword of damage x, this sword disappears. Then, if c_i gt 0, you will obtain a new sword with damage max(x, c_i); otherwise, you gain nothing.

Now you want to know the maximum number of monsters you can kill. Note that you can kill each monster at most once.

Input

Each test contains multiple test cases. The first line contains the number of test cases T ($1 \leq T \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and m ($1 \leq n,m \leq 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$).

The third line of each test case contains m integers $b_1, b_2, \ldots, b_m$ ($1 \leq b_i \leq 10^9$).

The fourth line of each test case contains m integers $c_1, c_2, \ldots, c_m$ ($0 \leq c_i \leq 10^9$).

It is guaranteed that the sum of n and m over all test cases does not exceed $2 \cdot 10^5$, respectively.

Output

For each test case output one integer — the maximum number of monsters you can kill.

Example

Input

```
5
3 2
2 2 2
2 3
3 2
2 3
2 3
2 3 4
0 0 0
3 5
1 7 7
6 6 2 2 2
2 0 0 7 2
4 4
1 5 3 5
7 4 6 5
0 0 1 6
2 2
1 1000000000
1000000000 1
1000000000 0
```

Output

```
2
2
5
3
2
```

Note

Visualizer link

In the first test case, you can first kill monster #1 using sword #1, and obtain a new sword with damage max(2,3)=3. You can then use this sword to kill monster #2.

In the second test case, you can't obtain any new swords because all c_i=0, so you can only kill monster #1 and #2 with your two existing swords.